

CSMVessel0000000007 Cruise Ship Resilience_ Propulsion Navigation

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | Date: June 2026 Category: Maritime / Propulsion

Revision Drivers (all V2.0 documents)

- 1. 2025-2026 experimental and regulatory data supersedes V1.0 specifications
- 2. Solar Cycle 25 SSN ~137 peak (observed) — elevated urgency for all Carrington-focused work
- 3. Standards updates — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Pod housing material	Steel/Al nacelle	BFRP nacelle (ClassNK 2024 structural equivalency)	Metal pod housing couples GIC and has no recycling path
GPS resilience	Not addressed	LoRa maritime + INS + Doppler backup: GPS-free navigation	GPS-denied operation during Carrington SEP events unaddressed in V1.0
DP system wiring	Copper CANbus	PMMA optical fiber ring bus (CSMFAB013 marine)	CANbus copper is primary GIC signal corruption path for DP
Shaft bearing isolation	Plastic/rubber shaft seal	Si3N4 ceramic stern tube bearing (R>10^12 Ohm)	V1.0 isolation material fails at temperatures generated by localized GIC heating
LNG integration	Generic propulsion	LNG-electric hybrid + LiFePO4 emergency: updated 2025 fleet data	V1.0 did not account for LNG fleet configuration specifics

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.